Interpreting Remainders in Real Situations
Answer Key

Grades 4–5 Mathematics

Quick Answers

1. 7
2. 5
3. 4
4. (a) estimated number of classrooms = 24, (b) classrooms that get a full set = 23
5. 41
6. (a) vans needed = 13, —
7. (a) whole pieces cut = 14, (b) centimetres of ribbon left = 10 cm
8. (a) estimated number of cartons = 10, (b) cans left over = 16
9. (a) markers per classroom = 46, (b) markers left over = 2
10. (a) completely full pallets = 13, (b) pallets that must be ordered = 14, —

What is verified

8 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 6, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 4–5 Mathematics
4.NBT.B.6 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 17 tagged checks.

Common wrong answers

3. *If they got 3: dropped the remainder instead of adding one more bus*
6. *If they got 12: stopped at the whole-number quotient and left 10 riders behind*

1. A camp counselor has 63 water bottles to load into coolers. Each cooler holds exactly 8 bottles. How many coolers can she fill completely?

Solution: Divide 63 by 8. Since $8 \times 7 = 56$ and $63 - 56 = 7$, the quotient is 7 with remainder 7.

The question asks for coolers that are *completely* filled, so the 7 leftover bottles do not make another full cooler. Drop the remainder. $\boxed{\text{7 coolers}}$

2. A baker makes 95 dinner rolls and puts 6 rolls in each bag. She fills as many bags as she can. How many rolls are left over?

Solution: Divide 95 by 6. Since $6 \times 15 = 90$ and $95 - 90 = 5$, the quotient is 15 with remainder 5.

Here the question asks about what is *left*, not how many bags. The remainder itself is the answer. $\boxed{\text{5 rolls}}$

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3. A total of 138 students are going to the science museum. Each bus seats 40 students. How many buses must the school order so that every student has a seat?

Solution: Divide 138 by 40. Since $40 \times 3 = 120$ and $138 - 120 = 18$, the quotient is 3 with remainder 18.

Three buses seat only 120 students, and 18 students would be left standing on the sidewalk. Every student needs a seat, so one more bus is required: round the quotient up. $\boxed{\text{4 buses}}$

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4. A school office has 476 pencils. Each classroom is given a set of 20 pencils. (a) Estimate the number of classrooms by first rounding 476 to the nearest ten. (b) Now find exactly how many classrooms can be given a full set of 20 pencils.

Solution (a): Round 476 to the nearest ten: 480. Then $480 \div 20 = 24$, because $20 \times 24 = 480$. The estimate is about 24 classrooms. $\boxed{24}$

Solution (b): Now divide exactly: $20 \times 23 = 460$ and $476 - 460 = 16$, so the quotient is 23 with remainder 16. Only 23 classrooms can get a *full* set of 20; the 16 remaining pencils are not enough for a 24th set. $\boxed{\text{23 classrooms}}$

The estimate of 24 was one too high, which is what a rounded-up dividend should do — a good check, not an exact answer.

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5. A librarian is shelving 1,000 books. Each shelf holds 24 books. How many shelves will be completely filled?

Solution: Divide 1,000 by 24. Building up: $24 \times 40 = 960$, and $1{,}000 - 960 = 40$, which still holds one more shelf: $24 \times 41 = 984$, leaving $1{,}000 - 984 = 16$.

So the quotient is 41 with remainder 16. Sixteen books are not enough to fill a 42nd shelf, so only 41 shelves are completely filled. $\boxed{\text{41 shelves}}$

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6. A field trip takes 154 riders. Each van carries 12 riders. (a) How many vans are needed so that everyone gets a ride? (b) Explain why the trip leader cannot order the number of vans that the plain quotient gives.

Solution (a): Divide 154 by 12. Since $12 \times 12 = 144$ and $154 - 144 = 10$, the quotient is 12 with remainder 10. Twelve vans carry 144 riders, so 10 riders would have no seat. Add one more van. $\boxed{\text{13 vans}}$

(b) — instructor-judged. Look for a response that says the remainder of 10 is 10 real riders who still need transport, so a 13th van must be added even though it will not be full. A response that only repeats “the answer is 13” without naming the leftover riders is incomplete.

Common wrong answer: 12 vans — the quotient reported without asking what happens to the remainder.

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7. A roll of ribbon is 500 centimetres long. Each bookmark uses a piece 35 centimetres long. (a) How many whole bookmark pieces can be cut? (b) How much ribbon is left over?

Solution (a): Divide 500 by 35. Since $35 \times 14 = 490$ and $500 - 490 = 10$, the quotient is 14 with remainder 10. Only whole pieces can be cut, so drop the remainder. $\boxed{\text{14 pieces}}$

Solution (b): The remainder is the ribbon that is left, measured in the same unit as the roll. $\boxed{\text{10 cm}}$

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8. A food drive collected 296 cans. The volunteers pack 28 cans into each carton. (a) Estimate the number of cartons by rounding both numbers to the nearest ten. (b) After every full carton is packed, how many cans are left over?

Solution (a): Round to the nearest ten: $296 \to 300$ and $28 \to 30$. Then $300 \div 30 = 10$, so about 10 cartons. $\boxed{10}$

Solution (b): Exactly: $28 \times 10 = 280$ and $296 - 280 = 16$, so the quotient is 10 with remainder 16. Ten cartons are packed full and the remainder is what stays on the table.

$\boxed{\text{16 cans}}$

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9. A school orders 8 boxes of markers with 144 markers in each box. The markers are then shared equally among 25 classrooms. (a) How many markers does each classroom receive? (b) How many markers are left over?

Solution: First find the total: $8 \times 144 = 1{,}152$ markers.

(a) Divide 1,152 by 25. Since $25 \times 46 = 1{,}150$ and $1{,}152 - 1{,}150 = 2$, the quotient is 46 with remainder 2. Sharing equally means every classroom gets the same whole number of markers. $\boxed{\text{46 markers}}$

(b) The 2 markers that cannot be shared equally are the remainder. $\boxed{\text{2 markers}}$

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10. A charity has 2,000 meal kits to ship. One pallet holds 144 kits. (a) How many pallets will be completely full? (b) How many pallets must the charity order so that every kit ships? (c) Explain why the answers to (a) and (b) are different.

Solution: Divide 2,000 by 144. Since $144 \times 13 = 1{,}872$ and $2{,}000 - 1{,}872 = 128$, the quotient is 13 with remainder 128.

(a) Thirteen pallets are packed to capacity; the 128 remaining kits are not enough to fill a fourteenth. $\boxed{\text{13 pallets}}$

(b) Those 128 kits still have to ship, so they need a pallet of their own — a partly loaded one. $\boxed{\text{14 pallets}}$

(c) — instructor-judged. Look for a response that contrasts the two questions: (a) counts only pallets that are completely full, (b) counts every pallet that leaves the warehouse, and the 128 leftover kits are what make the two counts differ by one. A response that says only "one is bigger" is incomplete.